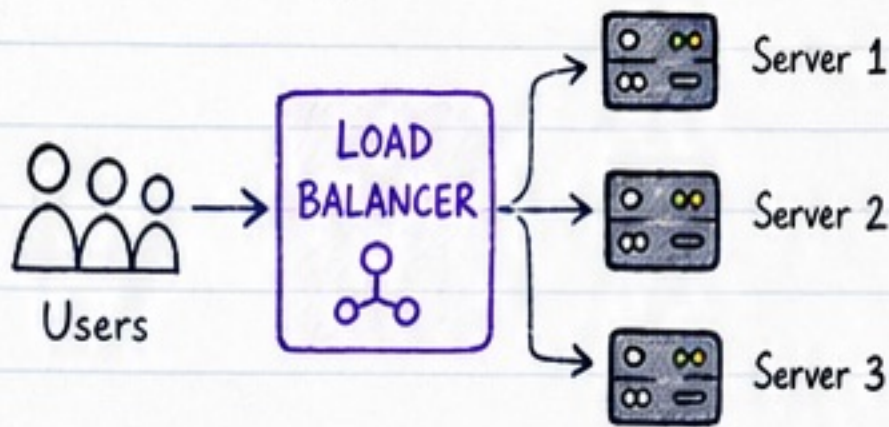


LOAD BALANCERS – QUICK NOTES



1) WHAT IS A LOAD BALANCER?

A Load Balancer sits between users and servers. It distributes incoming traffic across multiple servers to improve performance, reliability and availability.

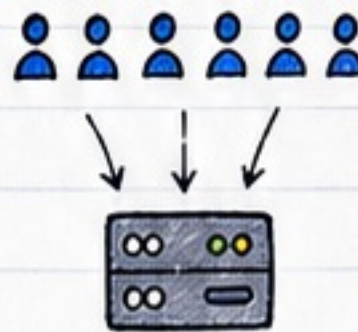


One entry point. Many servers.
Smart distribution.

2) WHY WE NEED IT

Without LB

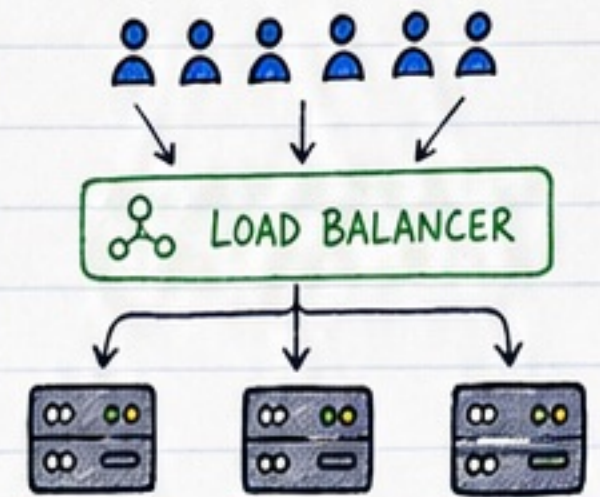
All traffic hits a single server.



Server Overload
Slow, Unreliable,
Single Point of Failure

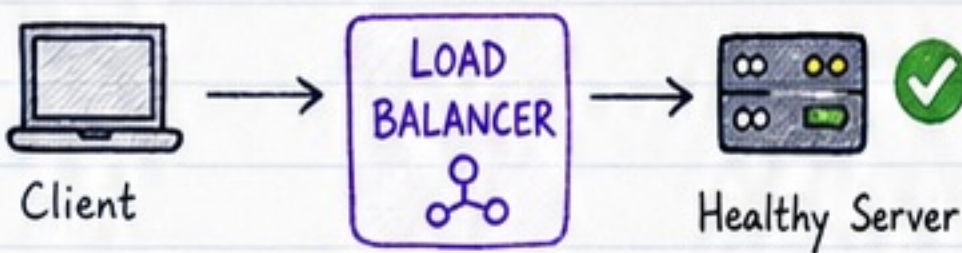
With LB

Traffic is distributed across servers.



High Availability
Scalable, Reliable, Fault Tolerant

3) BASIC FLOW



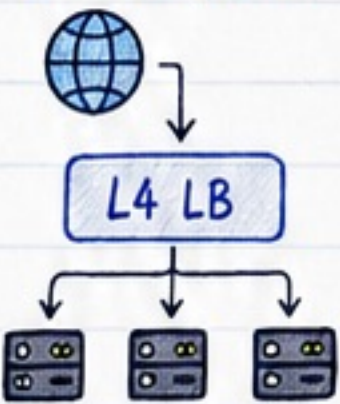
Client → Load Balancer → Healthy Server

4) LOAD BALANCING ALGORITHMS

- 1 Round Robin — sequential
- 2 Least Connections — fewest active
- 3 IP Hash — sticky routing
- 4 Weighted Round Robin — capacity based
- 5 Least Response Time — fastest healthy server

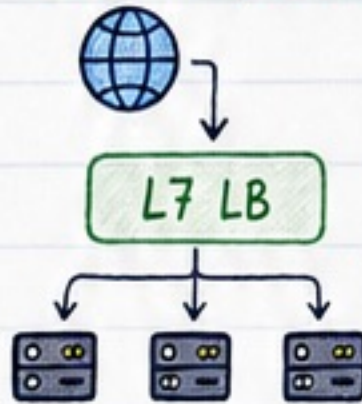
5) L4 vs L7 LOAD BALANCERS

L4: TCP/UDP



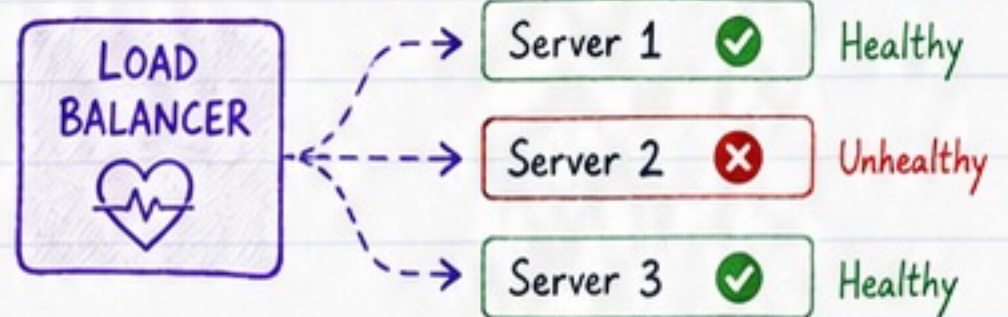
- Works on IP + Port (TCP/UDP)
- Faster
- Does not understand application data
- Used for TCP/UDP traffic

L7: HTTP/HTTPS



- Understands HTTP headers, paths, cookies, etc.
- Can do content-based routing
- Used for HTTP/HTTPS traffic

6) HEALTH CHECKS



- Health check: `/health` → 200 OK
- Unhealthy servers leave rotation
- Only healthy servers receive traffic
- When healthy again, added back automatically

7) LISTENER → ROUTING RULE → TARGET GROUP

LISTENER

Listens for incoming traffic on a protocol and port.

Example:

- HTTP : 80
- HTTPS : 443
- TCP : 3306

ROUTING RULE

Decides where to send the request.

Example:

- IF Path = `/api/*` → TG API
- IF Host = `api.example.com` → TG API
- IF Path = `/admin/*` → TG ADMIN

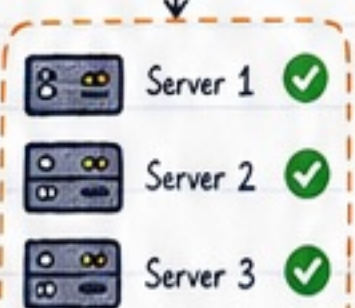
TARGET GROUP

Group of backend servers that receive traffic.

Example:

- TG PROD → Server 1, Server 2
- TG ADMIN → Server 3

TARGET GROUP (TG PROD)



Listener → Rule → Target Group → Server

8) FAILURE FLOW



Traffic is automatically routed to healthy servers.
No downtime for users.



GOLDEN RULE

Never send traffic to an unhealthy server.



WATCH OUT!

The load balancer must not become a single point of failure.

INTERVIEW TIP

Prefer stateless servers. Keep sessions in Redis or a database.



MEMORY HOOK

One entry point.
Many servers.
Smart distribution.

